QBism: Personalist Bayesian probability in quantum mechanics

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Imperial College, June 2026

2002: Quantum Bayesianism (not yet QBism)

PHYSICAL REVIEW A, VOLUME 65, 022305

Quantum probabilities as Bayesian probabilities

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(Received 15 August 2001; published 4 January 2002)

In the Bayesian approach to probability theory, probability quantifies a degree of belief for a single trial, without any *a priori* connection to limiting frequencies. In this paper, we show that, despite being prescribed by a fundamental law, probabilities for individual quantum systems can be understood within the Bayesian approach. We argue that the distinction between classical and quantum probabilities lies not in their definition, but in the nature of the information they encode. In the classical world, *maximal* information about a physical system is *complete* in the sense of providing definite answers for all possible questions that can be asked of the system. In the quantum world, *maximal information is not complete and cannot be completed*. Using this distinction, we show that any Bayesian probability assignment in quantum mechanics must have the form of the quantum probability rule, that maximal information about a quantum system leads to a unique quantum-

2010: QBism

QBism, the Perimeter of Quantum Bayesianism

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This article summarizes the Quantum Bayesian [1–7] point of view of quantum mechanics, with special emphasis on the view's outer edges—dubbed QBism.[1] QBism has its roots in personalist Bayesian probability theory, is crucially dependent upon the tools of quantum information theory, and most recently, has set out to investigate whether the physical world might be of a type sketched by some false-started philosophies of 100 years ago (pragmatism, pluralism, nonreductionism, and meliorism). Beyond conceptual issues, work at Perimeter Institute is focussed on the hard technical problem of finding a *good* representation of quantum mechanics purely in terms of probabilities, without amplitudes or Hilbert-space operators. The best candidate representation involves a mysterious entity called a symmetric informationally complete quantum measurement. Contemplation of it gives a way of thinking of the Born Rule as an *addition* to the *rules* of probability theory, applicable when an agent considers gambling on the consequences of his interactions with a newly recognized universal capacity: dimension (formerly Hilbert-space dimension). (The word "capacity" should conjure up an image of something like gravitational mass—a body's mass measures

to outward appearances is nearly identical to a common yearly annoyance. There are lessons here for quantum mechanics. In the history of physics, there has never been a healthier body than quantum theory; no theory has ever been more all-encompassing or more powerful. Its calculations are relevant at every scale of physical experience, from subnuclear particles, to table-top lasers, to the cores of neutron stars and even the first three minutes of the universe. Yet since its founding days, many physicists have feared that quantum theory's common annoyance—the continuing feeling that something at the bottom of it does not make sense—may one day turn out to be the symptom of something fatal.

There is something about quantum theory that is different in character from any physical theory posed before. To put a finger on it, the issue is this: The basic statement of the theory—the one we have all learned from our textbooks—seems to rely on terms our intuitions balk at as having any place in a fundamental description of reality. The notions of "observer" and "measurement" are taken as primitive, the very starting point of the theory

David Mermin, Nature, 27 March 2014

Pablo Picasso, *Le Vieux Marc* (oil on canvas), 1912.

QBism puts the scientist back into science

“QBism”: now officially a noun

Accepted Paper

QBism’s account of quantum dynamics and decoherence

Phys. Rev. A

John B. DeBrota, Christopher A. Fuchs, and Rúdiger Schack

Accepted 13 September 2024

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QBism:

In quantum mechanics, all probabilities are an agent's personal degrees of belief. They should not be thought of as representing ignorance, but as guiding the agent's decisions.

Three tenets of QBism

The world is indeterministic: Even wavefunctions do not evolve deterministically.

Agents play a fundamental role in the world.

The future is genuinely open.

QBism is a valid interpretation of quantum mechanics

QBism is consistent.

QBism is empirically adequate.

Some reasons to adopt QBism

- Quantum “nonlocality”
- No-go theorems
- The shape of the quantum formalism
- Extended Wigner’s friend scenarios
- The measurement problem

The measurement problem

AI Overview

The measurement problem in quantum mechanics refers to the conflict between two ways quantum systems evolve: the smooth, deterministic, and unitary evolution described by the Schrödinger equation (which leads to superpositions of states) and the sudden, probabilistic, and discontinuous "collapse" of a wave function into a single, definite outcome when a measurement is performed.

Mainstream attempts to “solve” the measurement problem

- Bohmian Mechanics
- Many Worlds
- Decoherence

Wigner's friend: a dramatic version of the measurement problem

Wigner's friend measures a qubit in her lab. She sees an outcome.

Wigner uses the Schrödinger equation to describe the lab and all its contents, including friend and qubit.

Collapse, or no collapse?

QBism's radical solution

Both quantum states (i.e., wavefunctions) and measurement outcomes are personal to the agent performing the measurement.

Quantum measurement

The mainstream approach:

A measurement is modeled by joint (Schrödinger) time evolution of a system and a meter, followed by a readout of the meter. The outcome is objective.

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QBism:

A measurement is an action an agent takes on a system. The meter is an extension of the agent. Outcomes as well as outcome probabilities are personal to the agent.

Measurement device = extension of the agent

Probabilities

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Like quantum states, probabilities of quantum measurement outcomes are objective and epistemic, i.e., they represent knowledge.

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What is objective are the rules of quantum mechanics and probability theory.

Personalist probability (de Finetti, Ramsey, Savage)

- The rules of probability theory are grounded in decision theory ("how should I act").
- They have a normative character.
- They can be derived from the requirement of "no sure loss" (Dutch book coherence).

Decision-theoretic personalist probability

My probability of an event E is p if I regard \$$p$ as the fair price of a lottery ticket that pays \$1 if E occurs.

Dutch book (adapted from Wikipedia)

horse	**odds offered**				
1 2 3	even 1:2 1:3				

Dutch book (adapted from Wikipedia)

horse	**odds offered**		**amount bet**		
1	even		\$120		
2	1:2		\$80		
3	1:3		\$60		
	total		\$260		

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Unlike roulette, where one is certain to lose in the long run, here the bettor will lose $20 with certainty in a single race!

Dutch book (adapted from Wikipedia)

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1	even	$1/2$	\$120	\$240	\$20
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Dutch book coherence

Definition

An agent's betting odds are called *Dutch book coherent* if they rule out the possibility of a Dutch book.

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How should I gamble?

The Dutch-book derivation results in a theory with a normative character.

Frequentism

Frequencies are data – Probabilities measure uncertainty

Frequencies can be assigned probabilities. Probabilities can be refined on the basis of measured frequencies.

But can't we measure probabilities?

Strictly speaking, we can only measure frequencies. But, as we saw:

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Key assumption: **Repeated Trial**

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Key assumption: **Repeated Trial**

Exchangeability characterises repeated trials

A potentially infinite sequence of trials is exchangeable if outcome probabilities don't depend on the order of the trials. If a sequence of trials is exchangeable, we can think of each trial as having the same unknown probability (de Finetti, 1928).

Repeated trials in quantum mechanics

Exchangeability characterises repeated trials in quantum mechanics

A potentially infinite sequence of trials is exchangeable if outcome probabilities don't depend on the order of the trials. If a sequence of trials is exchangeable, we can think of each trial as having the same unknown quantum state. This is the quantum de Finetti theorem (Caves, Fuchs, Schack, 2001).

Even in quantum mechanics, the notion of a repeated trial depends on an agent's prior probability (or prior state) assignments.

Thank you.

What are agents?

Informal, preliminary definition:

Agents are entities that

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Users of quantum mechanics are agents

- capable of applying the quantum formalism normatively.

Example: radioactive decay

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In QBism, the probability that an individual atom in a trap decays within the next day is not objective, because it depends on prior assumptions including the prior quantum state of the nucleus.